TheONE AI

Voice-First AI Companion • Unified Comedy Architecture

Live App: theone-ai.onrender.com
GitHub: github.com/Tarak098/TheONE-AI
Platform: Render Web Service (FastAPI Full-Stack)

1. EXECUTIVE OVERVIEW

TheONE AI is a minimal, voice-first companion inspired by ChatGPT's Advanced Voice Mode. It replaces traditional dense text chat with an audio-reactive **Central Voice Orb**. It operates under a single **Unified Comedy Identity**: ALWAYS funny, sarcastic, witty, and entertaining regardless of user mood. It enforces a strict **one-sentence limit** (≤ 15 words) to maximize comedic impact and eliminate filler.

2. END-TO-END VOICE PIPELINE

```
[ User Microphone ] ──► (16 kHz Linear PCM Mono WAV)
        │
        ▼
[ Gnani Prisma STT v3 ] ──► Fast BCP-47 Speech Recognition
        │
        ▼
[ Intelligence Orchestrator ]
 ├── Unified Comedy Persona (Always funny, sarcastic, witty)
 ├── Adaptive Comedy Styles (Witty, roast, situational, teasing)
 ├── Google Gemma 4 26B (Gemini 2.5 Flash Failover)
 └── Strict Response Validator (≤ 1 Sentence, 3–10 Words)
        │
        ▼
[ Gnani Timbre TTS v2.5 ]
 ├── Voice: "Yashvi" (Female, Cheerful 1.04x speed prosody)
 └── Voice: "Zayan" (Male, Deadpan 0.98x speed prosody)
        │  (48 kHz Mono WAV Audio)
        ▼
[ Web Audio API Player ] ──► Real-Time AnalyserNode ──► Voice Orb
```

3. UNIFIED COMEDY & ADAPTIVE HUMOUR

TheONE AI has removed all mode selectors. It automatically adapts its comedic style (witty sarcasm, savage comebacks, situational observations, wordplay, gentle teasing) while preserving its singular persona:

Canonical Examples Across User Moods:
- *Boredom:* "Even boredom is bored of you."
- *Failure:* "Your exam paper won this round."
- *Anger:* "Should I inform your patience department?"
- *Celebration:* "Finally, your talent escaped captivity!"
- *Confusion:* "Your brain is buffering; let's simplify it."
- *Sensitive topics:* Gentle, non-insensitive humor without breaking character.

4. CORE CAPABILITIES

- **Central Voice Orb:** 5-state Web Audio visualizer (`IDLE`, `LISTENING`, `THINKING`, `SPEAKING`, `ERROR`) reacting to microphone and speech waveform volume.
- **Independent Voice Selection:** 1-click toggle between **Yashvi** (Female) and **Zayan** (Male) in top navigation bar, saved to `localStorage`.
- **Strict 1-Sentence Constraint:** Maximum 1 sentence (≤ 15 words). No paragraphs, no generic corporate assistant tone.
- **Contextual Relevance:** Responses strictly target user concepts; factual queries answered accurately with humor.
- **Seamless Scrolling UI:** Global side scrollbar guarantees full accessibility to the chat field on all viewport heights.

5. PRODUCTION TECH STACK

Component	Technology
Backend	FastAPI 0.110+, Python 3.11, Uvicorn ASGI
Frontend	HTML5, CSS3, Vanilla JS (ES6+), Web Audio API
STT Engine	Gnani Prisma v3 REST API (vachana.ai)
LLM Engine	Google Gemma 4 26B (Google GenAI SDK)
TTS Engine	Gnani Timbre v2.5 REST API (48 kHz WAV)
Deployment	Render Web Service (Full-Stack ASGI on HTTPS)

6. RENDER DEPLOYMENT SPECS

- **Architecture:** Single ASGI service mounts `/frontend` statically, eliminating CORS and proxy latency.
- **Build Command:** `pip install -r requirements.txt`
- **Start Command:** `uvicorn main:app --host 0.0.0.0 --port $PORT`
- **Health Check:** `GET /health` returns `{"status":"healthy"}`
- **Microphone Access:** Browser Web Audio & MediaRecorder work natively over Render's HTTPS origin.